

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

BarangayHealth — Community Health Monitoring System

Aligned with UN SDG 3: Good Health and Well-Being

Course: PC11 — System Analysis and Design | BSIT 2nd Year | 2nd Semester | AY 2025–2026

Field	Detail
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Project Title	BarangayHealth — Community Health Monitoring System
SDG Alignment	UN SDG Goal 3: Good Health and Well-Being
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1. PROJECT CHARTER

1.1 Project Background and Rationale

The Philippines faces persistent challenges in grassroots healthcare delivery. Barangay Health Centers (BHCs) often rely on paper-based records, making health monitoring of constituents error-prone, slow, and inaccessible. Missed immunizations, unmonitored chronic diseases, and delayed prenatal check-ups contribute directly to preventable illnesses and deaths — issues directly addressed by **UN SDG Goal 3: Ensure healthy lives and promote well-being for all at all ages**. BarangayHealth is a web-based Community Health Monitoring System designed to digitize and streamline the clinical workflow of Barangay Health Centers.

1.2 Project Objectives

- To provide a centralized digital repository for resident health records.
- To automate scheduling of immunizations, prenatal check-ups, and consultations.
- To generate automated health status reports and analytics for barangay health officers.
- To enable residents to view their health records and appointment schedules online.
- To reduce paper-based errors and improve data integrity in community healthcare.

1.3 SMART Success Criteria

Criterion	Description	Metric / Target
Specific	Digitize patient record management for BHCs	All health records stored in the system
Measurable	Reduce missed immunization rate	Track 100% of immunization schedules
Achievable	Implement within standard SDLC for 2nd-year BSIT	Solo deliverable within semester timeline
Relevant	Addresses SDG 3 — Good Health	Mapped to SDG 3, Target 3.8
Time-bound	Completed by end of 2nd Semester AY 2025–2026	Presented on May 18–20, 2026

1.4 High-Level Risk Register

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R-01	Internet connectivity issues at BHC	High	High	Offline-capable data entry with sync
R-02	Low digital literacy of health workers	Medium	High	Simple UI + onboarding guide
R-03	Data privacy / RA 10173 compliance	Medium	High	Role-based access + audit logging
R-04	Incomplete resident data during rollout	Medium	Medium	Allow partial records with flags
R-05	Scope creep beyond solo capacity	Low	Medium	Strict MoSCoW prioritization

2. STAKEHOLDER ANALYSIS

2.1 Stakeholder Register & Interest/Influence Matrix

Stakeholder	Role in System	Interest	Influence	Engagement Strategy
Barangay Administrator	Manages system settings, user accounts	High	High	Primary sponsor; weekly updates
Barangay Health Worker (BHW)	Records patient data, schedules	High	High	Key user; training and feedback sessions
Resident / Patient	Views records, receives notifications	High	Low	Usability testing; simplified portal
Municipal Health Office	Receives aggregate reports	Medium	Medium	Monthly report delivery
Department of Health (DOH)	Policy compliance body	Low	High	Compliance documentation

2.2 Negotiation Log (Summary)

The following negotiations were conducted during requirements elicitation interviews with simulated stakeholder representatives. Each item reflects a resolved conflict between initial requests and system feasibility.

Log #	Stakeholder	Initial Request	Resolution	Date
N-01	Barangay Admin	SMS notification to all residents	Email + in-app notification (SMS deferred to v2)	Apr 28, 2026
N-02	Health Worker (BHW)	Full electronic prescription module	Prescription notes only; no e-Rx (out of scope)	Apr 28, 2026
N-03	Resident	Mobile app (Android/iOS)	Mobile-responsive web app in v1	Apr 29, 2026
N-04	Municipal Health	Real-time dashboard feed to MHO portal	Monthly PDF report export in v1	Apr 29, 2026

2.3 Requirements Elicitation — Interview Protocol Summary

Interview Objectives:

- Identify current pain points in manual health record management.
- Understand daily workflow of Barangay Health Workers.
- Determine minimum viable features for practical deployment.

Key Findings from Raw Notes:

- BHWs spend 2–3 hours daily searching paper folders for patient records.
- Residents frequently miss immunization schedules due to lack of reminders.
- Admins cannot generate any reports without manually tallying paper forms.
- There is no way to track which residents have pending prenatal consultations.
- Privacy is a major concern — residents want their data protected.

3. SPECIFIC REQUIREMENTS

3.1 Functional Requirements (MoSCoW — 'Shall' Syntax)

FR-ID	Priority	Requirement (Shall Statement)	User Role(s)
FR-01	Must-Have	The system SHALL allow the Barangay Admin to create, update, and deactivate user accounts.	Admin
FR-02	Must-Have	The system SHALL allow BHWs to register new resident health profiles with personal and medical data.	BHW
FR-03	Must-Have	The system SHALL allow BHWs to record immunization events linked to a specific resident profile.	BHW
FR-04	Must-Have	The system SHALL automatically compute and display the next scheduled immunization date based on vaccine type.	BHW, Resident
FR-05	Must-Have	The system SHALL allow BHWs to record prenatal consultation visits and maternal health data.	BHW
FR-06	Must-Have	The system SHALL allow residents to log in and view their own health record summary.	Resident
FR-07	Must-Have	The system SHALL generate a Monthly Health Summary Report (PDF) with immunization coverage and consultation statistics.	Admin
FR-08	Should-Have	The system SHALL send in-app notifications to residents for upcoming appointments.	Resident
FR-09	Should-Have	The system SHALL provide a dashboard with real-time counts of active cases, pending immunizations, and prenatal visits.	Admin, BHW
FR-10	Could-Have	The system SHALL allow export of reports to CSV format for MHO submission.	Admin
FR-11	Won't-Have	The system SHALL NOT include e-prescription generation in v1.	N/A

3.2 Non-Functional Requirements

NFR-ID	Category	Requirement
NFR-01	Security	The system SHALL implement role-based access control (RBAC) with encrypted authentication (bcrypt/JWT).
NFR-02	Privacy	The system SHALL comply with the Data Privacy Act of 2012 (RA 10173) — no PHI shared without consent.
NFR-03	Performance	The system SHALL load any page within 3 seconds on a standard 10 Mbps connection.
NFR-04	Usability	The system SHALL achieve a System Usability Scale (SUS) score of at least 70 in user testing.
NFR-05	Reliability	The system SHALL maintain 99% uptime during BHC operating hours (Mon–Sat, 8AM–5PM).
NFR-06	Scalability	The system SHALL support a minimum of 500 resident records per barangay without performance degradation.

NFR-ID	Category	Requirement
NFR-07	Maintainability	The system SHALL follow MVC architecture to allow modular updates and code maintenance.

4. LOGICAL USE CASE MODELS

4.1 Use Case Diagram Description (UML 2.5)

The Use Case Diagram below identifies all actors and system interactions. Three primary actors interact with the BarangayHealth system: **Barangay Administrator**, **Barangay Health Worker (BHW)**, and **Resident**. An optional fourth actor, **Municipal Health Office (MHO)**, interacts only with the report export feature.

Actors:

Actor	Description
Barangay Administrator	Manages system users, views system-wide reports and dashboards.
Barangay Health Worker (BHW)	Primary data entry user; registers residents, records health events, schedules appointments.
Resident / Patient	Authenticated beneficiary who views personal health records and upcoming appointments.
Municipal Health Office (MHO)	External actor who receives exported health reports (CSV/PDF).

4.2 Use Case Narratives — Must-Have Features

UC-01: Register Resident Health Profile

Field	Details
Primary Actor	BHW
Secondary Actors	Admin, System
Precondition	BHW is logged into the system.
Main Flow	1. BHW navigates to 'Residents' > 'Add New Resident'. 2. BHW fills in: Full Name, Date of Birth, Sex, Address, Contact Number, Blood Type, Allergies. 3. BHW submits the form. 4. System validates inputs and saves the profile. 5. System displays confirmation with the assigned Resident ID.
Postcondition	Resident profile is saved and searchable.
Alternative Flow	If required fields are missing, system displays validation error and does not save.

UC-02: Record Immunization Event

Field	Details
Primary Actor	BHW
Secondary Actors	System
Precondition	Resident profile exists in the system. BHW is authenticated.
Main Flow	1. BHW searches for the resident by name or ID. 2. BHW selects 'Immunizations' tab in the resident profile. 3. BHW selects the vaccine administered (e.g., BCG, Hepatitis B, MMR). 4. BHW enters the date administered and lot number. 5. System saves the record and automatically calculates the next due date. 6. System highlights overdue vaccines in red on the profile dashboard.

Field	Details
Postcondition	Immunization event is logged; next due date is auto-computed.
Alternative Flow	Vaccine type not in list: BHW may use 'Other' with manual entry.

UC-03: Generate Monthly Health Summary Report

Field	Details
Primary Actor	Admin
Secondary Actors	System
Precondition	Admin is authenticated with Admin privileges.
Main Flow	1. Admin navigates to 'Reports' > 'Monthly Summary'. 2. Admin selects the target month and year. 3. System aggregates: total consultations, immunization coverage %, new registrants, prenatal visits. 4. System generates a PDF report with charts and tables. 5. Admin downloads the PDF.
Postcondition	PDF report is generated and available for download.
Alternative Flow	No data for selected period: system displays 'No records found' notification.

5. SYSTEM BOUNDARY — CONTEXT DIAGRAM (LEVEL -1 DFD)

5.1 External Entities and Data Flows

The Context Diagram (Level -1 DFD) defines the outermost boundary of the BarangayHealth system. It uses Gane-Sarson notation and shows all external entities and the data flows between them and the central system process bubble.

External Entity	Data Flow TO System	Data Flow FROM System
Barangay Administrator	Login credentials, report parameters, user mgmt commands	Dashboard summary, reports, system alerts
Barangay Health Worker	Resident data, immunization records, appointment details	Resident profiles, immunization schedules, confirmation messages
Resident / Patient	Login credentials, appointment confirmations	Health record summary, appointment reminders
Municipal Health Office	Report export request (out-of-band trigger)	Exported PDF/CSV health reports
Email/Notification Server	(Internal subsystem trigger)	Appointment reminder emails to residents

5.2 System Boundary Statement

The BarangayHealth system boundary encompasses the following modules: (1) User Authentication & RBAC, (2) Resident Profile Management, (3) Immunization Tracking & Scheduling, (4) Prenatal Care Monitoring, (5) Consultation Record Management, (6) Report Generation Engine, and (7) Notification Service. Explicitly outside the boundary: Electronic Prescriptions, Pharmacy Management, Hospital Billing, and Direct DOH System Integration (deferred to future versions).